Report Covering the creation and implementation of the website needed for the CSCU9B3 PHP assignment

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***Abstract*—This document is a report detailing:**
The Use case Architecture Design Database Design Implementation Testing Data Management Considerations
of the Event Horizon website (neat name huh ?)

I. Use Case

A. Booking a Technology Workshop

Primary Actor: Registered User

Goal: To reserve a place at an available technology workshop trough the Event Horizon website.

Preconditions:

- The user has an existing account
- The user has access to the Event Horizon website
- At least one workshop is available to book

Trigger: The user chooses an event they wish to attend.

B. Main Success Scenario

1. The user visits the website
2. The user navigates to the login page and enters their credentials
3. The website validates the credentials and grants access
4. The user navigates to the events page where the events details are displayed
5. The user selects and event that has available spaces
6. The user chooses the option to book the selected event
7. The website confirms that the booking has been successful
8. The user can navigate to their bookings page to view their reservation

C. Additional User Outcome

After a successful reservation the user can access the My Bookings page to review their bookings and if they no longer wish to attend they can cancel the reservation, which is then reflected in the booking no longer showing up to that one user and a space becoming available on the Events page.

II. Architecture Design

The website uses a three tier architecture consisting of a web browser, and Apache web server running PHP and MySQL relational database. The browser provides the user interface using HTML and CSS, allowing users to log in, browse events, and manage bookings. When a user performs an action the browser sends a HTTP request to the Apache server, where PHP processes the request and communicates with the database using SQL queries. The database retrieves or modifies relevant data and returns the results to PHP, which then generates the appropriate HTML response for the browser. This architecture keeps the UI, application logic and data storage separate.



Fig. 1. The entity relationship diagram used to make the connections between tables

III. Database design

A. Database Overview

The websites use a MySQL database called event _horizon. It contains three tables; users, EVENTS and bookings. Each table has a specific purpose, allowing the website to store user accounts, the bookings made by users and available events.

B. Entities and Attributes

The users table contains the data for each registered account, this includes the unique user ID, their username and password hash (please note for the sake of this assignment the password has is the same for all users, in a real website implementation there would be a hash generator tied to the users table along with a sign up option, but that was not a requirement here so i did not add one for the sake of not making this assignment take even longer), along with their email address (also note none of the email address are real and neither are the users, they are all fictional characters, and one is even a fictional planet, again for the sake of simplicity and to not expose any real

user data). Passwords are stored as hashes for better security as well, rather than plain text.

The EVENTS table stores details about the workshops, this includes, dates, capacity, description and the events name. This allows the website to dynamically update its capacity and display that information to the users, rather than hard coding it using HTML.

And lastly the bookings table keeps track of per user reservations, containing the users ID, the events ID and the booking date, along with the booking ID and when the booking was created. This allows for each user to only see their own bookings and for there to be no double bookings.

C. Primary and Foreign Keys

Each table has its own primary key, which is used to identify individual records. The user_id primary key is in the users table, event_id is the primary key for the events table and booking_id is the primary key for the bookings table.

Moreover the bookings table also contains 2 foreign keys, thus, linking each booking to the suer who made it and the event they selected. This allows a user to make multiple bookings, while allowing an event to have multiple users booking it.

There also is a unique constraint applied to the user _id and event _id to prevent users booking the same event twice. Foreign key constraints also ensure the bookings remain linked to valid users and events, with associated bookings being deleted if the referenced records are removed.

D. How the Database Supports the Website

The database provides the info needed for the features of our website. When a user visits the events page, the website pulls the data from the database and calculates the remaining space for each one.

When a user attempts to make a booking, the system checks that wether the event is still able to accept more bookings and that the users hasnt already booked that event. If these conditions are met then the new booking is stored and tied to the user and the database updates the available spaces.

Users can also view their current bookings and cancelt them if they wish when said booking is cancelled its record is removed and the website recalculates available spaces to show other users that there is 1 extra free slot.

IV. Implementation

V. Testing

VI. Data Management Considerations